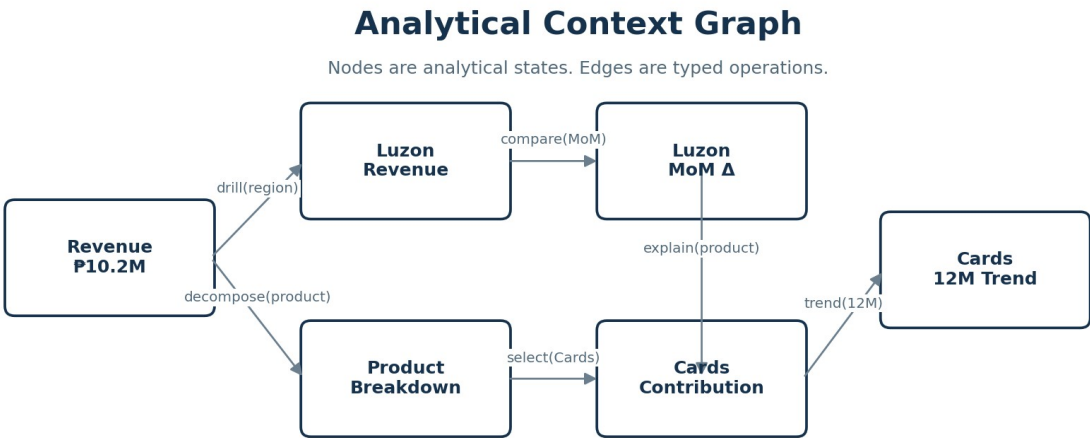


INTERACTIVE ANALYTICAL DASHBOARD FRAMEWORK

Design Plan

From static charts and filters to an explorable graph of analytical contexts



A dashboard becomes an explorable graph rather than a fixed collection of charts.

Concept document • September 2026

1. Executive Summary

The proposed framework treats a dashboard not as a collection of charts linked by filters, but as an interactive analytical workspace. Every visual mark - a bar, point, row, cell, or KPI - represents a well-defined analytical context that can be drilled, explained, compared, branched, trended, or decomposed.

Core design thesis

A chart is a view over an analytical node. User interactions create new nodes. Analytical operations form the edges between them. The resulting investigation is a graph that can be navigated, revisited, branched, and narrated.

Design goals

- Make every visual element an entry point into deeper analysis rather than a passive display.
- Separate analytical meaning from visualization so the same node can be rendered as a chart, table, KPI, or narrative.
- Allow multiple kinds of drill: structural hierarchy, alternative dimensions, and alternative metrics.
- Preserve investigation history and branches instead of replacing the previous state on every click.
- Keep numerical truth deterministic. AI may choose, explain, and narrate operations, but computation remains owned by the analytical engine.
- Use semantic metadata to determine which analytical operations are valid at any point.

Primary abstraction

Analytical Context Node

Node = Metric + Scope + Grain + Time + Comparison + Provenance

Edge = Typed analytical operation such as drill, compare, explain, trend, decompose, or branch.

2. Product Principles

Principle	Meaning
Context before chart	A visualization is only one representation of a richer analytical state.
Interactivity as grammar	Interactions are reusable analytical verbs, not chart-specific event handlers.
Branch, do not destroy	A user can keep alternative hypotheses alive and move between them.
Deterministic numbers	Aggregations, deltas, contributions, anomalies, and comparisons are computed by code.
Semantic safety	The framework blocks nonsensical aggregations or invalid grain changes based on metadata.
Explainable provenance	Every result can trace back to the context, operation, source data, and calculation.

What this is not

- Not merely cross-filtering between charts.
- Not a fixed Country → Region → Province drill path.
- Not an LLM that freely writes SQL or performs arithmetic without constraints.
- Not a charting library replacement; Vega or another renderer can remain the presentation layer.

What success feels like

A user sees a surprising bar, clicks it, asks “why?”, follows the strongest explanation, branches into a competing hypothesis, compares the two paths, pins the useful finding, and exports the investigation as a narrative - without manually rebuilding filters or opening separate dashboards.

3. The Analytical Context Model

Every selectable visual mark should resolve to an explicit context object. This becomes the contract between the UI, the semantic model, the analytical engine, and any AI layer.

Field	Example	Purpose
dataset	sales_monthly	Identifies the governed data source.
metric	revenue	Defines what is being measured.
scope / filters	country=PH; region=Luzon	Defines the selected population.
grain	region	Defines the current level of aggregation.
time	Aug 2026	Defines the observation period.
comparison	MoM	Defines a reference period or peer set.
provenance	dashboard/revenue_by_region/ bar=Luzon	Tracks how the context was reached.

Suggested context schema

```
Context(  
  dataset="sales",  
  metric="revenue",  
  filters={"country":"PH", "region":"Luzon"},  
  grain=["region"],  
  time="2026-08",  
  comparison="mom",  
  lineage=[...]  
)
```

Important implication

A click does not directly “change the chart.” A click creates or modifies a Context. The system then decides which analytical operations are legal and which views can represent the resulting node.

4. Analytical Grammar

The framework should expose a small vocabulary of reusable analytical verbs. These operations should be independent of chart type.

Verb	Meaning
Drill down	Move to a lower structural hierarchy, e.g., Region → Province.
Drill up	Return to a parent grain or previous node.
Decompose	Explain the same value through another dimension, e.g., Luzon revenue by Product.
Compare	Compare against prior period, peer, benchmark, or selected node.
Trend	Keep scope fixed and show history over time.
Contribution	Show each child's share of a parent value.
Change contribution	Show which children explain a MoM/QoQ/YoY change.
Distribution	Reveal the underlying population beneath an aggregate.
Exceptions	Surface unusual, extreme, or structurally different children.
Switch metric	Keep population fixed but change the measure.
Correlate	Inspect relationships with another metric where semantically valid.
Branch	Create a separate investigation path without destroying the current one.
Pin	Keep a finding or node visible for later comparison or reporting.

Three distinct forms of drill

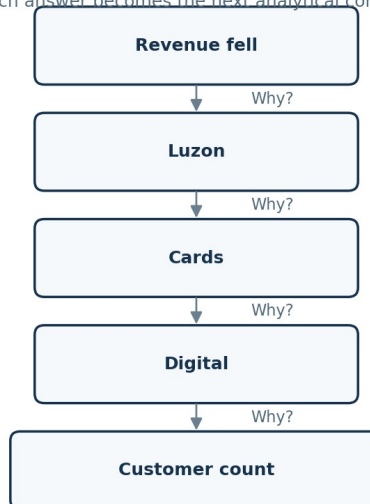
Type	Example	Question answered
Structural drill	Region → Province → City	What sits below this hierarchy level?
Dimensional drill	Luzon → Product / Channel / Customer	What alternative breakdown explains this number?
Metric drill	Revenue → Volume → Margin	What else is true about this same population?

5. “Explain This” and the Why-Chain

The signature interaction can be a universal “Explain” command available on any valid visual mark. The system should compute candidate explanations deterministically and present the strongest paths for exploration.

Interactive “Why?” chain

Each answer becomes the next analytical context.



Candidate explanation signals

- Absolute contribution to the parent value.
- Contribution to period-over-period change.
- Concentration: whether a small number of children dominate the result.
- Offsetting contributors: large values moving against the parent direction.
- Deviation from historical baseline or peer distribution.
- New, missing, or structurally unusual contributors.
- Metric relationships, where explicitly allowed by semantic metadata.

6. Investigation Trees and Branching

Traditional drill-down replaces the previous view. The proposed framework should preserve prior analytical states, allowing the user to keep multiple hypotheses active.

Interaction	Behavior
Filter	Apply the selected scope to connected views.
Focus	Highlight the selection while retaining the full comparison population.
Branch	Open a new analytical path beside the current path.
Pin	Freeze a node or finding for later comparison.
Merge / compare	Bring two branches into a common comparison view.

Example investigation

Revenue decline → Region → Luzon → Product → Cards → Channel → Digital

A second branch could simultaneously explore Customer Segment. The interface should make the path visible as breadcrumbs or an investigation map rather than hiding it inside filter state.

Why branching matters

- Supports competing explanations instead of forcing a single linear path.
- Makes comparison between hypotheses natural.
- Preserves cognitive context and prevents users from getting lost after many drills.
- Creates a reusable investigation object that can later be shared or narrated.

7. Analytical Lenses

A higher-level abstraction can group verbs into “lenses.” A lens expresses the intent of the analysis and maps to a deterministic family of operations.

Lens	User intent	Typical operations
Composition	What makes this number?	decompose, contribution, drill
Change	What caused the movement?	delta, change contribution, offsets
Comparison	Why is A different from B?	peer compare, variance decomposition
Trend	How did we get here?	time series, inflection detection, rolling comparison
Exception	What looks unusual?	outlier, threshold, historical deviation
Relationship	What moves with this?	correlation / regression where valid
Distribution	What is underneath this aggregate?	histogram, quantiles, record-level view

UX advantage

A user does not need to know the exact analytical operation. They can choose an intent such as “Change” or “Composition,” and the framework proposes valid next actions from the current context.

Suggested next questions

Because the semantic model knows the current metric, grain, hierarchies, and comparisons, the UI can generate context-sensitive prompts such as:

- What makes up this value?
- Why is this category larger than its peer?
- How has this changed over the last 12 months?
- Which children offset the parent movement?
- Is this observation unusual relative to history?
- Which alternative dimension best explains the change?

8. Semantic Model and Governance

High interactivity only remains trustworthy if the framework knows what operations make sense. A semantic model should describe dimensions, hierarchies, metrics, grain, formulas, valid aggregations, time semantics, and relationships.

Metadata area	Examples
Dimensions	country, region, province, product, channel, customer_segment
Hierarchies	country → region → province → city
Metric type	additive, semi-additive, ratio, stock, rate, count
Aggregation	sum, mean, weighted mean, distinct count, recompute formula
Valid grain	the dimensions and time levels where the metric is meaningful
Comparison semantics	MoM, QoQ, YoY, plan vs actual, peer benchmark
Relationships	allowed joins, parent-child mappings, alternative metric links
Formatting	currency, percentage, basis points, units, scale
Permissions	row / column / metric visibility and role constraints

Example metric rule

margin_pct

formula = margin / revenue

aggregation = recompute_from_components

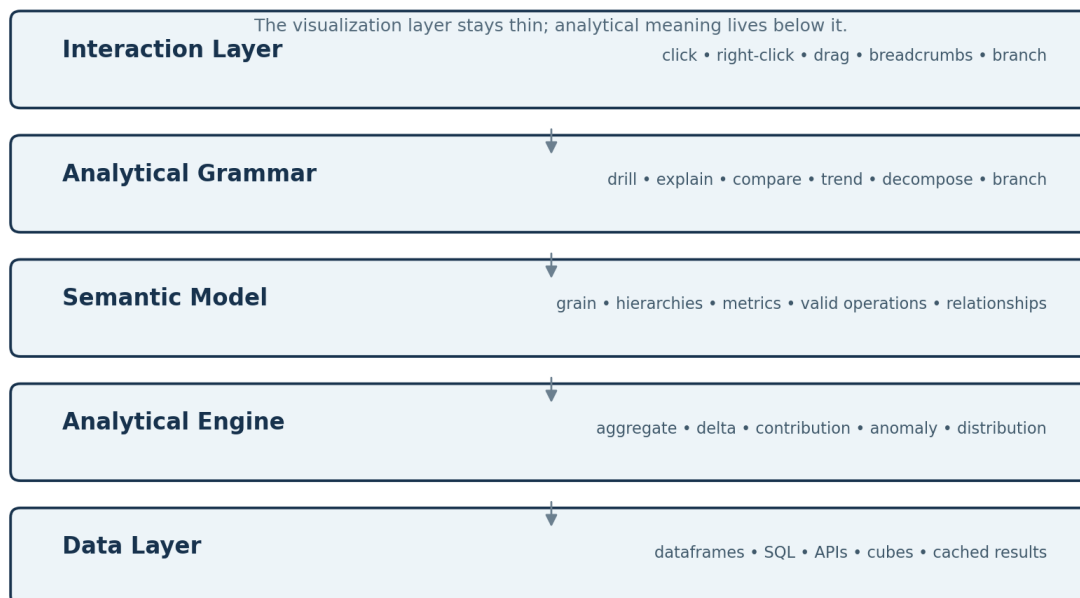
invalid = sum(margin_pct)

valid = aggregate margin and revenue, then divide

This metadata should be used both to validate operations and to generate the menu of possible next steps. Invalid analyses should be blocked before any chart or narrative is generated.

9. System Architecture

Five-layer architecture



Responsibilities

Layer	Responsibilities
Interaction	Capture mark selection, context menus, drag, focus, branch, breadcrumbs.
Analytical grammar	Translate UI intent into typed operations over context nodes.
Semantic model	Validate grain, metrics, dimensions, hierarchies, formulas, permissions.
Analytical engine	Execute deterministic computation and return result handles + metadata.
Data layer	Read governed datasets and execute filtered / grouped queries efficiently.
AI assistant (sidecar)	Interpret questions, choose valid operations, narrate computed results, suggest next actions.
Renderer	Render analytical nodes as Vega charts, tables, KPIs, narratives, or maps of the investigation.

Recommended event loop

Interaction flow

Select mark → emit structured selection → resolve Context → advertise valid operations → execute deterministic operation → add graph node → render the resulting view.

10. Vega/Vega-Lite Interaction Strategy

Vega can remain the visualization layer while the analytical framework owns meaning and state. The chart should emit selection signals rather than implementing business logic internally.

Vega responsibility	Framework responsibility
Render marks, axes, tooltips, highlighting	Define the metric, grain, scope, and valid operations
Emit selected mark / datum	Convert datum into a Context node
Show focus or filtered state	Decide whether action is filter, focus, branch, or compare
Render a new spec from returned data	Compute the data and preserve investigation lineage
Support contextual UI hooks	Populate context menu from semantic capabilities

Suggested payload from a selected mark

```
{
  "view_id": "revenue_by_region",
  "mark": {"region": "Luzon", "revenue": 5400000},
  "context_id": "ctx_184",
  "gesture": "contextmenu"
}
```

Design constraint

Do not let each chart invent its own drill behavior. The same selected context should expose the same analytical grammar regardless of whether it was selected from a bar chart, table, KPI, or another supported view.

11. Role of AI

AI is valuable as an interaction and reasoning layer, but it should sit on top of governed analytical primitives rather than directly manipulating raw data without controls.

AI may	AI should not
Translate “Why did revenue fall?” into a valid explain operation.	Invent numerical drivers without analytical evidence.
Choose among candidate dimensions suggested by metadata.	Aggregate a ratio or metric at an invalid grain.
Summarize deterministic result tables.	Perform hidden arithmetic that cannot be reproduced.
Suggest next questions from the current context.	Bypass permissions or semantic constraints.
Turn an investigation tree into a narrative report.	Hide which operations and filters produced a conclusion.

Deterministic explanation contract

Pattern

AI request → typed analytical operation → deterministic result object → AI narrative

The narrative should carry references to the node IDs and result objects used to generate it.

12. Implementation Roadmap

A staged build is preferable. The framework should prove the context model and analytical grammar before investing in a large dashboard builder.

Phase	Scope	Exit criteria
0. Semantic prototype	One dataset; define metrics, dimensions, grain, hierarchies, time, valid operations.	Machine-readable metadata can validate basic aggregations.
1. Context + drill	Bar chart selection; Context object; structural and dimensional drill; breadcrumbs.	Any selected bar can produce a valid child view with lineage.
2. Compare + trend	MoM/QoQ/YoY compare, time trend, focus/filter modes.	A node can be compared and trended without manual query construction.
3. Explain engine	Contribution, change contribution, offsets, concentration, anomalies.	“Explain this” returns deterministic ranked evidence.
4. Branching graph	Preserve nodes, branch investigations, pin findings, compare branches.	Users can navigate and revisit investigation history.
5. AI sidecar	Natural-language intent, next-question suggestions, narrative generation.	AI only calls governed operations and cites result nodes.
6. Dashboard builder	Reusable components, saved layouts, reusable contexts, refreshable data.	Users can create and save interactive analytical workspaces.

Recommended MVP demo

- Dataset with Geography, Product, Channel, Revenue, Volume, Margin, and monthly time.
- Landing view: Revenue by Region.
- Right-click a region → Drill by Province / Explain / Trend / Compare / Branch.
- Explain → ranked product contributors including large offsetting contributors.
- Branch → open Channel explanation next to Product explanation.
- Pin one finding and generate a short narrative of the investigation path.

13. Proposed Core Objects and APIs

Object	Purpose
Context	Immutable analytical state.
MetricSpec	Aggregation, formula, formatting, valid grains.
DimensionSpec	Hierarchy, members, parent-child relationships.
Operation	Typed transformation from one Context to another.
ResultHandle	Reference to deterministic output table / series / statistics.
ViewSpec	Maps a result into a visualization or narrative representation.
InvestigationGraph	Stores nodes, edges, branches, pins, and history.
ExplanationResult	Ranked evidence for why / contribution / anomaly questions.

Illustrative API

```
ctx2 = drill(ctx1, dimension="province")
ctx3 = compare(ctx2, period="previous_month")
exp = explain(ctx3, candidate_dimensions=["product", "channel"])
ctx4 = select(exp, member="Cards")
ctx5 = trend(ctx4, months=12)
```

Architectural payoff

Because operations act on Context rather than charts, the same engine can support web dashboards, notebooks, reports, AI agents, and future visualization types without rewriting analytical logic.

14. Key Design Decisions to Resolve

Decision	Recommended starting position
Node mutability	Use immutable Context nodes; every operation creates a new node.
State storage	Client graph for active session + server persistence for saved investigations.
Data ownership	Analytical engine owns data values; UI and AI pass handles / contexts.
Operation discovery	Semantic model advertises valid capabilities for each node.
Chart system	Start with Vega-Lite for common views; use full Vega only where interaction requires it.
Explanation ranking	Begin with contribution/change/offset/concentration rules before ML-based ranking.
AI role	Tool caller and narrator; never the primary numerical computation engine.
Performance	Cache result handles by context hash; push filters/aggregations to the data layer.
Auditability	Persist node lineage, operations, metric definitions, source version, and timestamps.

Immediate next build artifact

The most useful next step is a thin vertical prototype, not a full dashboard designer. Build one dataset, one initial bar chart, one context menu, a Context object, three operations (drill, explain, trend), an investigation breadcrumb, and a deterministic contribution engine. Once that interaction feels natural, generalize the semantic model and branching graph.

North-star experience

Any meaningful visual mark can answer: “What is this?”, “What is underneath it?”, “Why did it move?”, “How does it compare?”, “What else should I inspect?”, and “How did I get here?”

End of design plan